

```
!pip install chromadb
!pip install sentence-transformers
!pip install langchain
```

```
Collecting chromadb
  Downloading chromadb-1.5.6-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.0 kB)
Collecting build>=1.0.3 (from chromadb)
  Downloading build-1.4.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.12.3)
Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
Collecting pybase64>=1.4.1 (from chromadb)
  Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl.metadata (5.0 kB)
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.5.0)
Collecting onnxruntime>=1.14.1 (from chromadb)
  Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.2 kB)
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.20.0)
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
  Downloading opentelemetry-exporter-otlp-proto-grpc-1.40.0-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.20.0)
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
Collecting pypika>=0.48.9 (from chromadb)
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.80.0)
Collecting bcrypt>=4.0.1 (from chromadb)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.24.1)
Collecting kubernetes>=28.1.0 (from chromadb)
  Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.1)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.8)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.26.0)
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (24.0)
Collecting pyproject_hooks (from chromadb)
  Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.12.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from chromadb) (2025.1.31)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.14.0)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (25.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2023.12.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.36.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.25.0)
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.17.0)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.9.0)
Requirement already satisfied: websocket-client!=0.40.0,!=0.41.*,!=0.42.*,>=0.32.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.7.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.32.3)
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.0)
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.3.0)
Collecting durationpy>=0.7 (from chromadb)
  Downloading durationpy-0.10-py3-none-any.whl.metadata (340 bytes)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.5.0)
```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/token)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%                               349/349 [00:00<00:00, 24.4kB/s]

config_sentence_transformers.json: 100%           116/116 [00:00<00:00, 9.94kB/s]

README.md:      10.5k/? [00:00<00:00, 571kB/s]

sentence_bert_config.json: 100%                   53.0/53.0 [00:00<00:00, 4.93kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_
config.json: 100%                                612/612 [00:00<00:00, 47.4kB/s]

model.safetensors: 100%                          90.9M/90.9M [00:01<00:00, 453MB/s]

Loading weights: 100%                            103/103 [00:00<00:00, 403.44it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%                       350/350 [00:00<00:00, 18.9kB/s]

vocab.txt:      232k/? [00:00<00:00, 9.22MB/s]

tokenizer.json:   466k/? [00:00<00:00, 17.1MB/s]

special_tokens_map.json: 100%                     112/112 [00:00<00:00, 11.1kB/s]

config.json: 100%                                190/190 [00:00<00:00, 15.9kB/s]

```

```

documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation.",
    "Neural networks are inspired by the human brain.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds hidden patterns.",
    "Reinforcement learning learns through rewards.",
    "AI is used in healthcare and finance.",
    "Chatbots use NLP techniques.",
    "Computer vision deals with image processing.",
    "Transformers are used in modern NLP models.",
    "Embeddings convert text into vectors.",
    "Vector databases store embeddings efficiently.",
    "Cosine similarity measures similarity between vectors.",
    "Large Language Models generate text.",
    "Data preprocessing is important in ML.",
    "Feature engineering improves model performance.",
    "Overfitting occurs when a model learns noise."
]

ids = [f"doc{i+1}" for i in range(len(documents))]

```

```
embeddings = model.encode(documents)
```

```

collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)

```

```

query = "What is Artificial Intelligence?"

query_embedding = model.encode([query])

results = collection.query(
    query_embeddings=query_embedding,
    n_results=3
)

print(results)

```

```
{'ids': [['doc1', 'doc2', 'doc10']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of
```

```

for doc in results['documents'][0]:
    print(doc)

```

```

Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.
AI is used in healthcare and finance.

```

```

def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(query_embeddings=query_embedding, n_results=3)
    return results['documents'][0]

print(retrieve("Explain machine learning"))

```

```
['Machine Learning is a subset of AI.', 'Supervised learning uses labeled data.', 'Deep Learning uses neural networks
```

```

def pretty_retrieve(query):
    print("Query:", query)
    print("\nTop Results:\n")

    results = retrieve(query)
    for i, doc in enumerate(results):
        print(f"{i+1}. {doc}")

pretty_retrieve("What is deep learning?")

```

```
Query: What is deep learning?
```

```
Top Results:
```

1. Deep Learning uses neural networks.
2. Machine Learning is a subset of AI.
3. Neural networks are inspired by the human brain.